

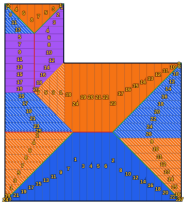
Flood Roofing — L-shape sweep (34/34 CLEAN)

Eight L proportions × four orientations (N/E/S/W) plus two mirrored handedness checks — 34 cases through the production hip engine. CLEAN = gap and overlap both under 1.5% (4px raster).

A1 — Square wing (canonical)

300×300 wing / 900×700 main · N (wing top-left)
70 sheets · gap 0.0% · overlap 0.0%

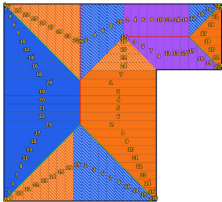
CLEAN



A2 — Square wing (canonical)

300×300 wing / 900×700 main · E (rot 90cw)
70 sheets · gap 0.0% · overlap 0.0%

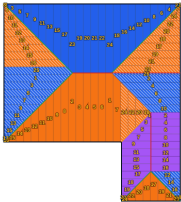
CLEAN



A3 — Square wing (canonical)

300×300 wing / 900×700 main · S (rot 180)
70 sheets · gap 0.0% · overlap 0.0%

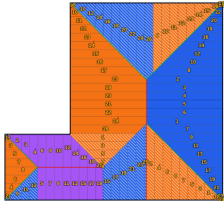
CLEAN



A4 — Square wing (canonical)

300×300 wing / 900×700 main · W (rot 270cw)
70 sheets · gap 0.0% · overlap 0.0%

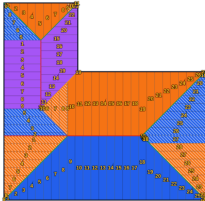
CLEAN



B1 — Wide-wing

356×329 wing / 972×615 main · N (wing top-left)
78 sheets · gap 0.0% · overlap 0.0%

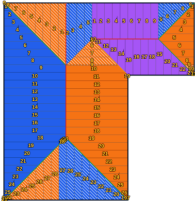
CLEAN



B2 — Wide-wing

356×329 wing / 972×615 main · E (rot 90cw)
78 sheets · gap 0.0% · overlap 0.0%

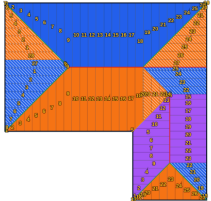
CLEAN



B3 — Wide-wing

356×329 wing / 972×615 main · S (rot 180)
78 sheets · gap 0.0% · overlap 0.0%

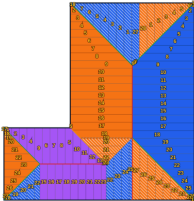
CLEAN



B4 — Wide-wing

356×329 wing / 972×615 main · W (rot 270cw)
78 sheets · gap 0.0% · overlap 0.0%

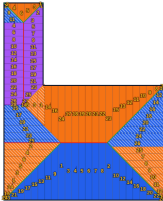
CLEAN



C1 — Narrow tall wing

220×460 wing / 900×640 main · N (wing top-left)
79 sheets · gap 0.0% · overlap 0.0%

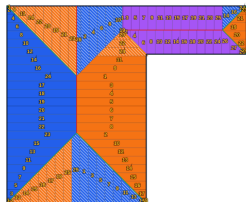
CLEAN



C2 — Narrow tall wing**CLEAN**

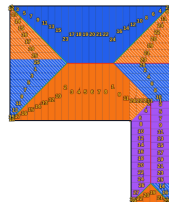
220×460 wing / 900×640 main · E (rot 90cw)

79 sheets · gap 0.0% · overlap 0.0%

**C3 — Narrow tall wing****CLEAN**

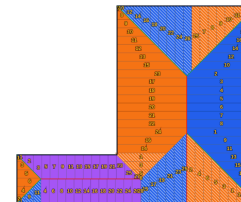
220×460 wing / 900×640 main · S (rot 180)

79 sheets · gap 0.0% · overlap 0.0%

**C4 — Narrow tall wing****CLEAN**

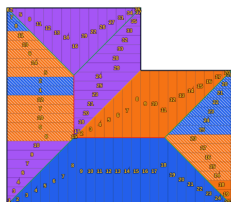
220×460 wing / 900×640 main · W (rot 270cw)

79 sheets · gap 0.0% · overlap 0.0%

**D1 — Fat short wing****CLEAN**

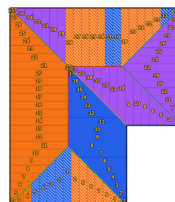
560×260 wing / 940×560 main · N (wing top-left)

83 sheets · gap 0.0% · overlap 0.0%

**D2 — Fat short wing****CLEAN**

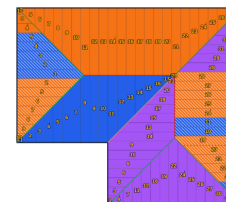
560×260 wing / 940×560 main · E (rot 90cw)

83 sheets · gap 0.0% · overlap 0.0%

**D3 — Fat short wing****CLEAN**

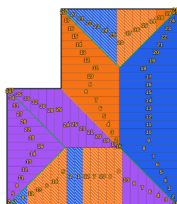
560×260 wing / 940×560 main · S (rot 180)

83 sheets · gap 0.0% · overlap 0.0%

**D4 — Fat short wing****CLEAN**

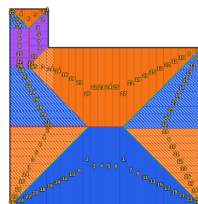
560×260 wing / 940×560 main · W (rot 270cw)

83 sheets · gap 0.0% · overlap 0.0%

**E1 — Small wing / big main****CLEAN**

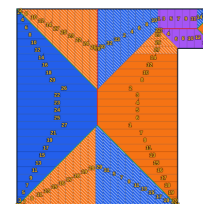
200×200 wing / 1000×820 main · N (wing top-left)

76 sheets · gap 0.1% · overlap 0.0%

**E2 — Small wing / big main****CLEAN**

200×200 wing / 1000×820 main · E (rot 90cw)

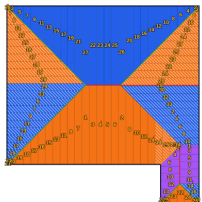
76 sheets · gap 0.1% · overlap 0.0%



E3 — Small wing / big main**CLEAN**

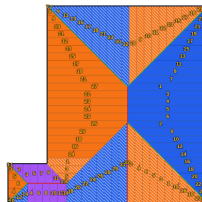
200×200 wing / 1000×820 main · S (rot 180)

76 sheets · gap 0.0% · overlap 0.0%

**E4 — Small wing / big main****CLEAN**

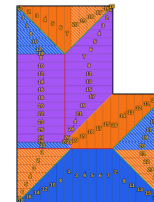
200×200 wing / 1000×820 main · W (rot 270cw)

76 sheets · gap 0.1% · overlap 0.0%

**F1 — Big wing / small main****CLEAN**

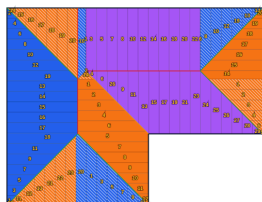
460×420 wing / 720×520 main · N (wing top-left)

72 sheets · gap 0.0% · overlap 0.0%

**F2 — Big wing / small main****CLEAN**

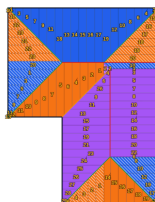
460×420 wing / 720×520 main · E (rot 90cw)

72 sheets · gap 0.0% · overlap 0.0%

**F3 — Big wing / small main****CLEAN**

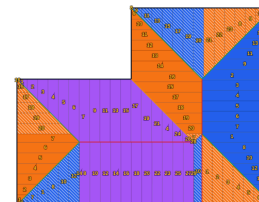
460×420 wing / 720×520 main · S (rot 180)

72 sheets · gap 0.0% · overlap 0.0%

**F4 — Big wing / small main****CLEAN**

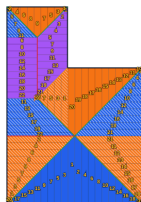
460×420 wing / 720×520 main · W (rot 270cw)

72 sheets · gap 0.0% · overlap 0.0%

**G1 — Near-square whole****CLEAN**

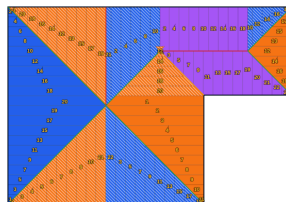
340×340 wing / 760×760 main · N (wing top-left)

66 sheets · gap 0.0% · overlap 0.0%

**G2 — Near-square whole****CLEAN**

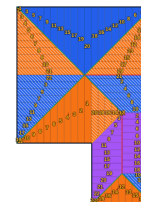
340×340 wing / 760×760 main · E (rot 90cw)

66 sheets · gap 0.0% · overlap 0.0%

**G3 — Near-square whole****CLEAN**

340×340 wing / 760×760 main · S (rot 180)

66 sheets · gap 0.0% · overlap 0.0%

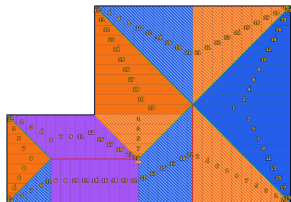


G4 — Near-square whole

CLEAN

340×340 wing / 760×760 main · W (rot 270cw)

66 sheets · gap 0.0% · overlap 0.0%

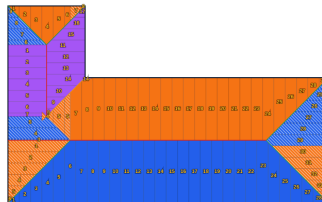


H1 — Long thin main

CLEAN

260×240 wing / 1080×420 main · N (wing top-left)

79 sheets · gap 0.0% · overlap 0.1%

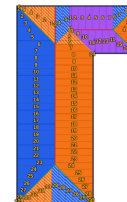


H2 — Long thin main

CLEAN

260×240 wing / 1080×420 main · E (rot 90cw)

79 sheets · gap 0.0% · overlap 0.1%

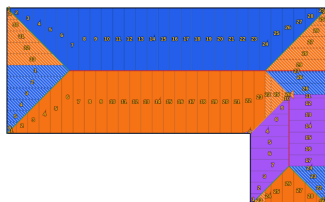


H3 — Long thin main

CLEAN

260×240 wing / 1080×420 main · S (rot 180)

79 sheets · gap 0.0% · overlap 0.1%

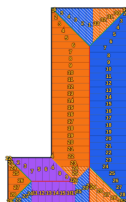


H4 — Long thin main

CLEAN

260×240 wing / 1080×420 main · W (rot 270cw)

79 sheets · gap 0.0% · overlap 0.0%

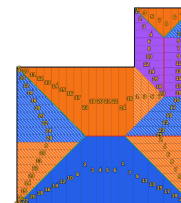


M1 — Square wing (canonical) (mirrored)

CLEAN

300×300×900×700 · horizontal mirror (notch top-right)

70 sheets · gap 0.0% · overlap 0.0%



M2 — Wide-wing (mirrored)

CLEAN

356×329×972×615 · horizontal mirror (notch top-right)

78 sheets · gap 0.0% · overlap 0.0%

